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Subsidies and market share in the global steel industry

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Abstract

This paper examines how subsidies affect competitive outcomes in the global steel industry. Using firm-level data from the OECD MAGIC database covering 2006–2022, the analysis assesses the relationship between government support and market-share developments across steel producers worldwide. Econometric results indicate that subsidies increase recipients' global market share at the expense of less-subsidised competitors. Firms receiving larger support through cash grants and below-market borrowings tend to gain market share even when they exhibit weaker productivity, cost efficiency and financial performance: subsidies weaken the normal link between firm performance and market-share gains. The analysis also identifies negative spillovers on competing firms, implying that support granted to some producers can reduce rivals' market shares and discourage investment by unsubsidised firms. The results are consistent across OECD Members and partner economies, although subsidisation levels are significantly higher in the latter and hence the dampening of market signals is even more pronounced there. Overall, the evidence suggests that subsidies contribute to resource misallocation, persistent steel excess capacity and shifts in global competitive positions. These findings highlight the importance of greater transparency in industrial support and stronger international cooperation to reduce distortions and support a more level playing field in the global steel sector.

Keywords: Steel industry; excess capacity; market share; industrial subsidies; cash grants; tax rebates; government ownership; distortions; below-market finance

JEL codes: H25 ; H32 ; L52 ; L61

Contact

Fabien Mercier (✉ fabien.mercier@oecd.org).

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This project was executed in close collaboration with the OECD Trade and Agriculture Directorate (TAD) and benefited significantly from the development of the **Manufacturing Groups and Industrial Corporations (MAGIC) database**. The expertise of TAD colleagues in estimating below-market borrowings – loans provided to steel firms on more favorable terms than a competitive market would allow – was invaluable. We especially thank **Jehan Sauvage** and **Yuki Matsumoto** for those contributions, as well as for their estimations of each steel firm's global market share.

The study was designed, conducted, and drafted by **Fabien Mercier** of the Directorate for Science, Technology and Innovation (STI). Additional colleagues who provided feedback on this study include **Anthony de Carvalho**, **Stephan Raes**, and **Jerry Sheehan** of STI. Their valuable comments and careful review of earlier versions of the paper, along with their sustained interest, were instrumental.

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Executive Summary

Governments continue to support steel producers through grants and below-market borrowings (BMB). Such support, often provided in large and non-transparent amounts in some partner economies (OECD, 2025[1]), risks tilting competition and locking in inefficient capacity by keeping uneconomic plants operating or encouraging new capacity additions not aligned with market conditions. Previous analyses by the Steel Committee have shown that higher levels of subsidisation in partner economies contribute to capacity increases.

We found:

- **Subsidies increase recipients' market share to the detriment of their competitors**. Firms that receive the most support tend to win global market share[1] at the expense of firms that receive little or none. Econometric regressions found negative spill-over of the subsidies provided to other firms on a firm's market share, and positive impact of cash grants, BMB and corporate tax rebates on a recipient's market share. Yet the productivity, cost efficiency and financial strength of subsidised firms do not improve, and are often lower than lesser subsidised firms to start with.
- **Normal market signals are blurred**. In healthy markets, higher profitability, higher capacity utilisation and a lower debt burden tend to be rewarded with larger sales volumes. When support increases or is larger (as in partner economies), these signals fade: highly subsidised firms gain ground even when they are less profitable, less cost-efficient and more indebted than their peers. The more a firm is subsidised, the lesser its market share gains correlate with its profitability, its cost-efficiency, its capacity utilisation and a lower debt burden.
- **Spill-overs hurt rivals**. When one firm is subsidised, competing firms lose ground: negative spill-overs (market share losses) to competitors are evidenced in our model estimations. Furthermore, subsidies may crowd out private investment to the sector at large and discourage rivals from expanding.
- **No broad efficiency gains**. Earlier OECD work shows that subsidies fail to raise productivity across sectors (OECD, 2025[2]). Combined with our new evidence on market-share shifts in the steel sector, this suggests that higher support, as is the case in partner economies, leads to lower overall welfare as resources are misallocated.
- **Patterns are global**. The blurring effect of subsidies on the normal correlations between market share outcomes and their market-drivers is visible in both OECD Member countries and partner economies. This is despite partner economies starting with much higher levels of subsidisation than OECD Member countries and hence much lower level of correlation between market share and its normal drivers.
- **Divergences in subsidisation levels play out consistently with those results:** in OECD Member countries, firms tended to lose market share through most of the study period (2006-2022), with the least subsidised OECD Member countries' firms losing the most. By contrast, in partner economies, the most subsidised firms gained the most market share in 2009: subsidies seem to aggressively increase market shares in partner economies, to the detriment of OECD Member countries.

How can subsidies boost market share without delivering better performance? Subsidies may:

- **Undercut prices**. Support may lower production costs, letting recipients sell below market prices and capture customers even when they are not more efficient.
- **Deter investments from competitors**. Persistent subsidisation may signal to competitors and potential entrants that future margins will be thin, reducing their incentive to modernise, expand capacity, or enter the market at all.
- **Demand diversion**. Subsidised firms may enjoy preferential access to public procurement or other demand-side measures, diverting orders their way regardless of their actual efficiency and performance.

Consequences may include:

- **Resource misallocation and excess capacity**. Capital, labour and innovation flow toward firms chosen as subsidy recipients rather than by performance, holding back sector-wide productivity and resulting in sub-optimal aggregate economic welfare. By keeping inefficient firms in the marketplace, subsidies also lead to excess capacity.
- **Trade tensions**. Uneven subsidisation stirs disputes and retaliation, undermining efforts to create a level playing field for trade and complicating co-operation on broader policy goals such as decarbonisation.
- **Fiscal risks**. Propping up weaker firms can become an open-ended commitment, diverting public funds from more productive uses, with possible dire reckoning further down the road.

Policy considerations for governments:

- **International cooperation that strengthens subsidy disciplines, enhances business certainty and reduces trade frictions would be preferable to unilateral action and can be expected to curb the use of such measures** - a main conclusion from the earlier joint work of the International Monetary Fund (IMF), the World Bank (WB), the World Trade Organization (WTO) and the OECD (IMF et al., 2022[3]).
- **Make a concerted effort to reduce subsidies to the steel sector in a coordinated manner - including with partner economies**. A rapid and uncoordinated phase-out in jurisdictions that are not among the most subsidy-intensive could unintentionally strengthen the market share of more heavily subsidised firms, even without an increase in their own support.
- **Make current subsidies more transparent and targeted to meet the intended policy goal**: poorly targeted support is more likely to weaken the link between firm performance and market outcomes. Greater transparency and alignment with clearly defined and publicly accepted policy goals can reduce distortions.
- **Subsidies that lower the marginal costs of production may distort price competition even more**, allowing recipient firms to undercut rivals and expand unfairly. This does not imply that other subsidies are more desirable, but subsidies that reduce steel prices act in a very direct way through trade and the global price channel and would explain a large part of the loss of market shares for non-recipients.
- **Be aware of overlapping support**: steel firms receiving grants, below-market finance or tax breaks often benefit from additional instruments like public procurement access. These effects are cumulative and must be assessed jointly by policy makers and analysts.
- **Preserve competitive neutrality**: subsidies should not weaken market signals that reward efficiency. Soft budget constraints - like persistent bailouts or countercyclical BMB - risk locking in inefficient capacity and deterring needed restructuring.

1 Introduction

Robust and data-driven empirical analysis is vital to inform policy debates on the impacts of subsidies and their policy implications. Building on established OECD analytical frameworks and the expertise developed under the Steel Committee's current Programme of Work and Budget (PWB), this report investigates how subsidies influence the international competitiveness of steel firms, with particular attention to their relationship with market share.

Policy concerns around subsidies, as well as financial and non-financial government support more broadly (hereafter "subsidies"), typically focus on their potential to distort market outcomes. Subsidies are likely to exacerbate global excess capacity across several strategically important sectors, including steel, shipbuilding, solar panels, and aluminium. By enabling selected firms to sustain production and capacity levels that would not be viable under normal market conditions, subsidies undermine a level playing field (OECD, 2025[4]). Over the years, there has been a strong and sustained call for greater transparency in the provision of subsidies (IMF et al., 2022[3]). A significant milestone in that respect was the establishment of the OECD MAGIC (MAnufacturing Groups and Industrial Corporations) database, which enabled firm-level studies across a number of sectors (Annex B).

In the steel industry, excess capacity remains a particularly acute problem (OECD, 2025[1]). Both subsidies and excess capacity can depress global steel prices and heighten competitive pressures on unsubsidised firms, eroding fair competition. Furthermore, subsidies are unevenly distributed across jurisdictions: for example, a typical Chinese steel firm receives 5 times more subsidies in the form of grants and below-market borrowings than a steel firm located in another partner economy, and 10 times more than a steel firm located in an OECD Member country (Mercier and Giua, 2023[5]). Consequently, subsidies risk increasing non-market driven exports and aggravating international trade tensions.

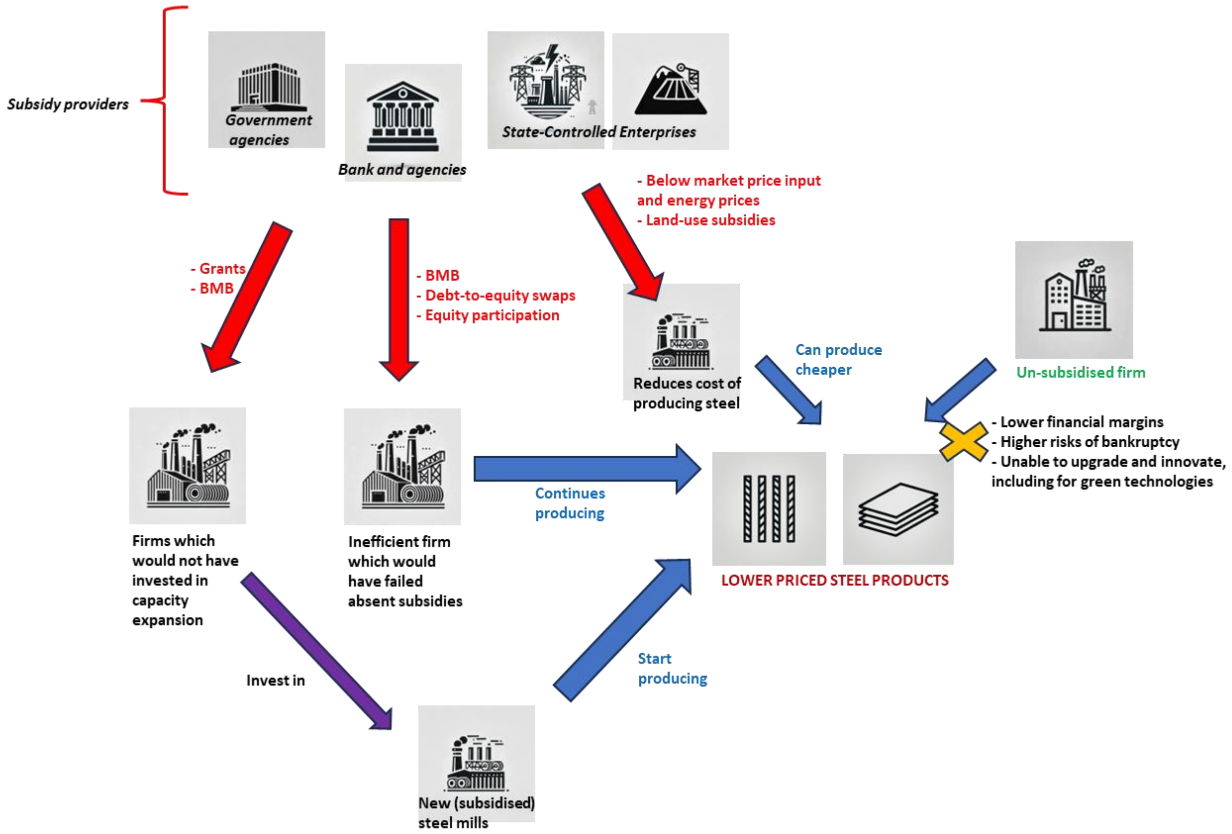
Concerns over subsidies typically fall into two categories. **The first category is structural: subsidies can keep uneconomic plants operating or encourage new capacity additions not in line with market considerations, thereby sustaining global excess capacity** (OECD, 2025[1]) (Mercier, 2024[6]). Such dynamics depress prices over the long term and threaten the viability of unsubsidised firms. Previous analyses by the Steel Committee found that grants and below-market financing have had a large and statistically significant impact in supporting both capacity maintenance and expansion. In partner economies, a grant worth USD 1 million annually, sustained over a number of years, was found to be associated with a capacity increase from anywhere between 5 000 to 15 000 metric tonnes. Moreover, a USD 1 million in subsidies through below-market borrowings (BMB) was found to increase capacity by about 1 000 metric tonnes outside of steel crises, while strong anecdotal evidence suggested it had a powerful counter-cyclical effect in impeding capacity closures during steel crises (OECD, 2025[7]). Hence, the effect of subsidies on capacity expansion and maintenance is an established fact, and its impact would be felt both in terms of higher production volumes and lower international prices (Figure 1).

The second category of concerns relates to the more general distortive effects of subsidies. Even in the absence of subsidy-induced capacity expansions, subsidies can distort competitive conditions both domestically and internationally. OECD analysis shows that subsidies generally flow toward firms that are larger, more indebted, and that have greater state ownership (Mercier, 2024[6]). This finding was further validated by broader OECD sectoral studies (OECD, 2024[8]), (OECD, 2025[2]).

Because subsidies of both categories may disrupt market outcomes and international competition, they can fuel trade tensions and result in retaliatory measures across sectors. Furthermore, if subsidies weaken market signals, then they can enable less efficient firms to maintain or even expand their market share to the detriment of less subsidised competitors, domestic or abroad, which would result in a clear sub-optimal allocation of resources and a net loss to aggregate welfare world-wide (Figure 1).

This report investigates the potential for subsidies to weaken market signals, and to help recipients acquire market shares, irrespective of their potential capacity expansions. It focuses on the link between subsidies and firm-level market share dynamics. The analysis draws on the OECD's MAGIC database, which combines information on government support to manufacturing firms with detailed financial data.

Figure 1. Subsidies impact both quantity and prices, reducing margins of unsubsidised firms



Note: Most common instruments used for each effect are listed in red near the red arrows representing the subsidisation incurred.
Source: OECD Secretariat.

Subsidies are often nested into more general industrial policies, which should embed continuous evaluation to minimise distortions (Criscuolo and Lalanne, 2022[9]). Hence, it is quite likely that subsidies operate through more numerous channels that the channels of quantities and prices depicted in Figure 1 above.

The remainder of this report is organised as follows. Section 2 presents key stylised facts on the relationship between subsidies and market share. Section 3 applies econometric analysis to examine these dynamics while controlling for other relevant factors, building on methods used in previous Steel Committee work (OECD, 2025[4]), but focusing exclusively on the steel sector. Section 4 concludes with a discussion of policy implications.

2 Subsidies weaken the relationship between market share and its drivers

This section presents descriptive evidence on the interplay between market share, subsidies, and other variables typically associated with market share dynamics. The aim is both to illustrate the strength of these potential relationships under different subsidisation regimes and to guide the selection of relevant explanatory variables for the "market share" variable in the econometric analysis of Section 3.

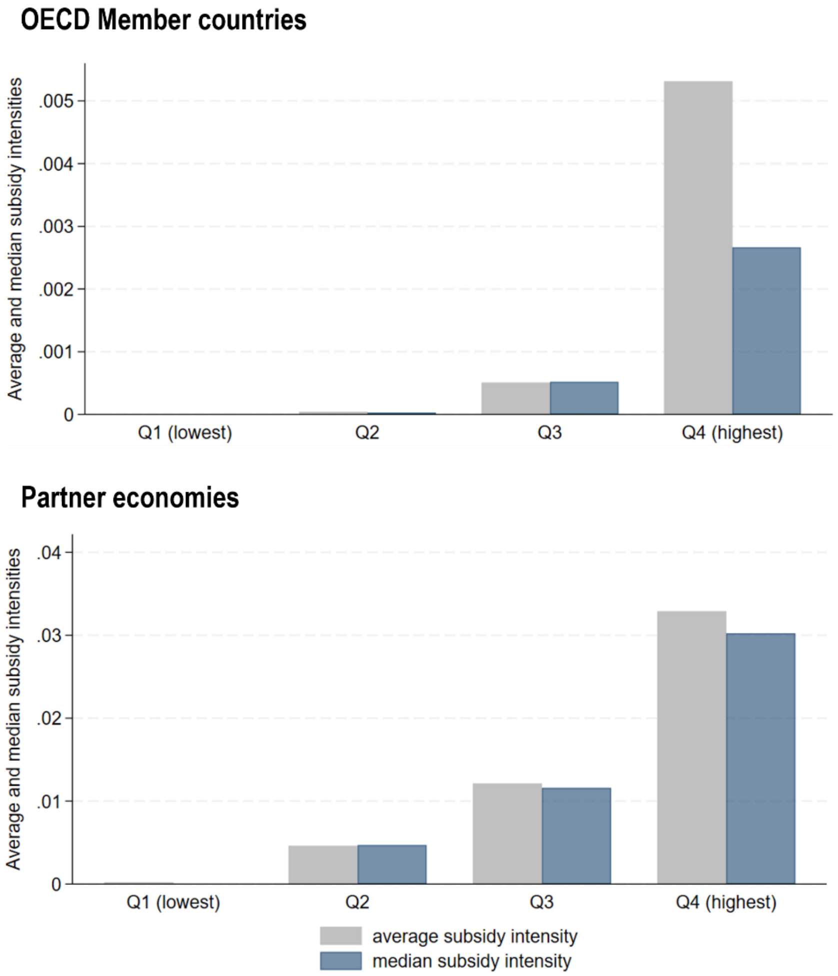
2.1. Subsidy intensities: definition and salient facts

Two common instruments to provide subsidies to steel firms are grants and below-market borrowings (BMB), though many other instruments exist (Mercier and Giua, 2023[5]). Grants are direct cash transfers, with amounts typically available from corporate filings at the firm level or from government websites at the programme level. BMB are estimated by the OECD by comparing the actual interest rates paid by firms with a benchmark reflecting normal market conditions and the borrowers' financial profiles, adjusted to exclude the effect of potential government guarantees that may artificially boost steel firms' credit ratings.

While many other subsidy instruments exist (Mercier and Giua, 2023[5]), this report focuses on grants, BMB, and, to a lesser extent, corporate income tax concessions, as support through other channels is far more difficult to quantify. To gauge the extent of subsidisation across firms of different sizes, the report uses the ratio of subsidies to total assets, referred to as "subsidy intensity". For example, in Section 3 the analysis employs ratios of grants, BMB, their combined total, and tax concessions to assets as regressors for changes in market share. In this section, however, we focus specifically on the combined grant and BMB-to-asset ratio.[2]

The results presented in this section remain valid even when the sample is restricted to firms in OECD member countries or to those in partner economies. This is notable given the significant differences in subsidy intensity between the two groups. Figure 2 illustrates these differences by showing the average and median subsidy intensities, by quartile, for each sub-sample. Notice how the vertical axes for OECD Member countries and partner economies are graduated (scale of subsidisation is 10 times less for OECD Member countries).

Figure 2. Partner economies subsidise 10 times more than OECD Member countries



Note: This is an observation-level graph, meaning that a single firm may appear in different subsidy intensity quartiles across years, depending on its annual subsidy-to-asset ratio. Subsidy quartiles are defined based on the 25% quantiles of the combined grants and below-market borrowings (BMB) relative to total assets - i.e. (grants + BMB) / assets.
Source: OECD MAGIC database for steel firms.

As a consequence of these very different levels of subsidisation, when quartiles are calculated for the full sample, steel firms from OECD Member countries are largely concentrated in the first and second quartiles, with 95 and 90 observations respectively, compared with only 17 and 5 in the third and fourth quartiles. Each quartile contains 174 observations in total, though 109 were excluded from the figure because of missing data for certain variables.

2.1.1. Market share gains, profitability, indebtedness and subsidies (aggregate for the period)

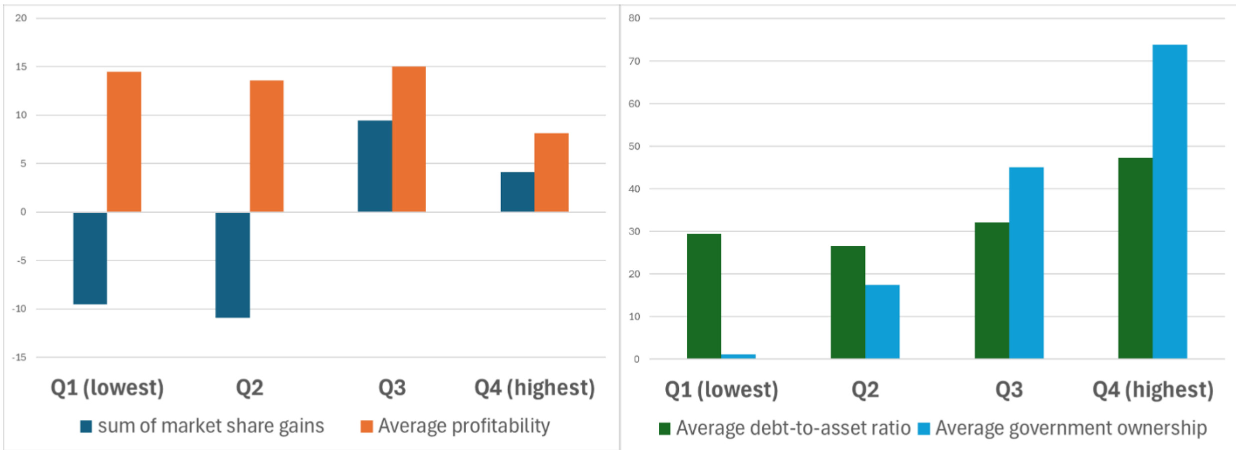
Figure 3 shows, by subsidy-intensity quartile, the total absolute market share gains of firms in each quartile, together with their average profitability (EBITDA-to-asset ratio), indebtedness (debt-to-asset ratio), and share of government ownership (in per cent). The subsidy-intensity quartiles Q1, Q2, Q3 and Q4 are displayed on the X axis, from the lowest quartile Q1 to the highest quartile Q4. Hence, subsidisation increases when one moves from left to right. Firms represented and assigned to each quartile are all the steel firms covered in the OECD MAGIC database (both in OECD Member countries and partner economies).

Firms in the lower subsidy-intensity quartiles (Q1–Q2) consistently lost market share over the period, despite having comparable or stronger profitability and lower indebtedness than firms in the higher quartiles (Q3–Q4). For example, Q4 firms were over 50% more indebted relative to assets and nearly 50% less profitable than Q1 firms, yet they gained market share while Q1 firms lost it.

The largest cumulative market share gains - nearly 10% - are observed in the third quartile (Q3), which combines relatively high subsidy intensities with moderate debt levels, solid profitability, and average government ownership of 45%. By contrast, gains in the most-subsidised quartile (Q4) were smaller, likely reflecting subsidies directed at distressed firms, where support intensifies as governments intervene to prevent bankruptcy. Strikingly, firms with higher debt-to-asset ratios are gaining market share, even though such ratios are typically associated with higher default probabilities in virtually every mainstream credit risk model (Mercier, 2016[10]).

Figure 3. Less-subsidised steel firms are losing market share to more heavily subsidised competitors, despite having stronger financial performance

Firms in lower subsidy intensity quartiles are losing market share, despite having overall higher average profitability and lower debt-to-asset ratio



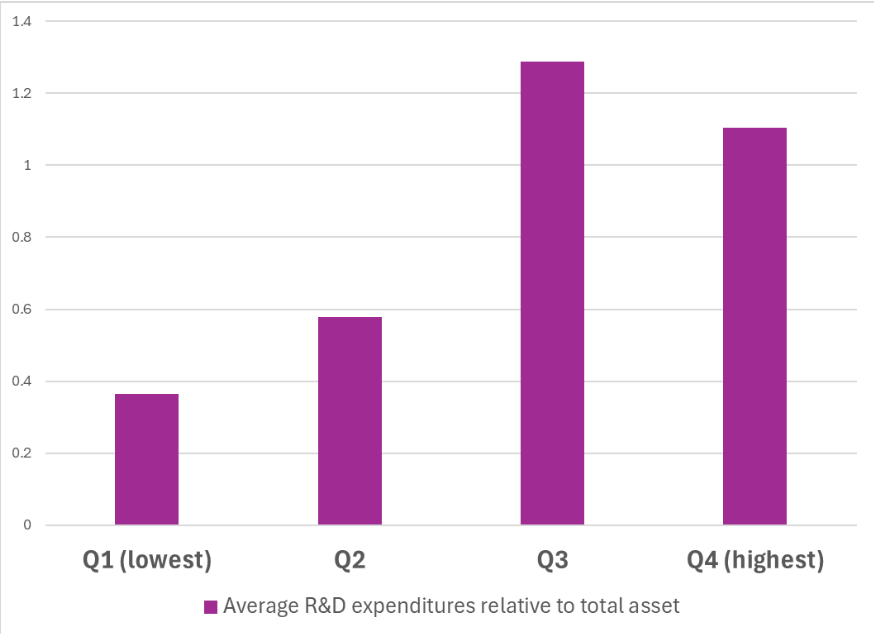
Note: 1) The subsidy-intensity quartiles are displayed on the X axis, from the lowest quartile Q1 to the highest quartile Q4. Hence, subsidisation increases when one moves from left to right.
2) This is an observation-level graph, meaning that a single firm may appear in different subsidy intensity quartiles across years, depending on its annual subsidy-to-asset ratio. Subsidy quartiles are defined based on the 25% quantiles of the combined grants and below-market borrowings (BMB) relative to total assets - i.e. (grants + BMB) / assets. By construction, each quartile contains approximately the same number of observations (around 174 per quartile).
Source: OECD MAGIC database for steel firms.

Capacity utilisation, the ratio of production to capacity, is broadly similar across quartiles, averaging around 80% (not shown in Figure 3). A potential theoretical explanation for this uniformity would be that it likely reflects the global nature of excess capacity: all producers are price takers, taking as given the steel price for each product and market, and adjusting their capacity utilisation accordingly. What differs is the level of subsidy support, which reduces effective production costs and cushions less efficient firms. As a result, capacity utilisation rates do not vary systematically by quartile: subsidies, by lowering effective costs and cushioning profit margins for less efficient firms, would allow them to keep running at similar capacity utilisation levels than more efficient firms. An equivalent phrasing is that subsidised firms produce more (than their market variables would allow in a free-market economies), while unsubsidised firms, faced with market share losses, cut back on capacity to remain profitable. The result is the observed similar capacity utilisation rates for all quartiles of the distribution.

R&D spending, measured relative to assets, increases with subsidy intensity. Yet for the most heavily subsidised firms (Q4), this higher spending does not appear to translate into greater profitability (Figure 4). Furthermore, Q4 firms actually spent less than Q3 firms on R&D in spite of higher subsidies. This again may be the effect of a counter-cyclical use of subsidies directed towards ailing firms (which would be over-represented in Q4), or simply of subsidies being used for other purposes than R&D, such as mere capacity expansions.

Figure 4. Average R&D expenditures to asset ratio, by subsidy intensity quartile

More subsidised firms tend to have higher R&D expenditures



Note: This is an observation-level graph, meaning that a single firm may appear in different subsidy intensity quartiles across years, depending on its annual subsidy-to-asset ratio.
Source: OECD MAGIC database for steel firms.

The preceding figures use annual firm-level data, with subsidy-intensity quartiles recalculated each year. Firms may therefore move between quartiles over time, depending on how their subsidy-to-asset ratio compares with that of their peers. This captures the shifting nature of subsidies: a firm may receive substantial support one year and none the next. To assess the medium-term performance of the most subsidised firms, however, a different approach is needed.

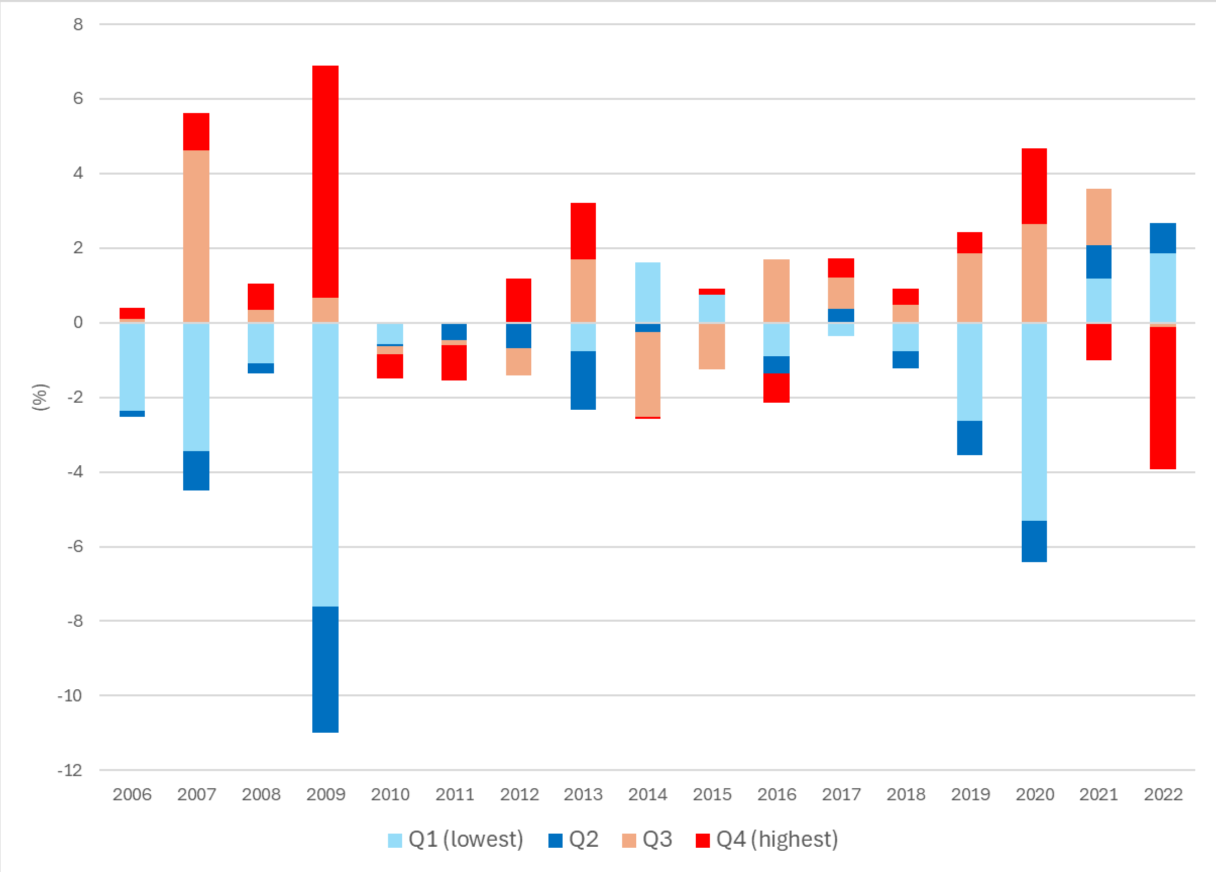
2.1.2. Subsidy intensity and market share developments over time

To avoid year-to-year reallocation across quartiles, in this section firms are classified according to their ***average*** subsidy-to-asset ratio ***over the entire period of examination***. This fixed classification ensures that a firm remains in the same quartile across all years, even if its annual subsidy levels fluctuate. For instance, a firm ranked in Q4 (highest subsidy intensity) on the basis of its multi-year average may receive no support in a given year, but is still treated as highly subsidised firm relative to its peers.

Figure 5 presents annual market share changes of steel firms by subsidy-intensity quartile. Firms in the higher-subsidy quartiles (Q3–Q4) generally show stronger market share growth than those in the lower quartiles, though this pattern breaks down in some years, notably 2021 and 2022.

Figure 5. Most subsidised firms tend to have better yearly gains than least subsidised firms (with exceptions in 2021 and 2022)

Yearly changes of market share per (fixed over time) subsidy intensity quartile



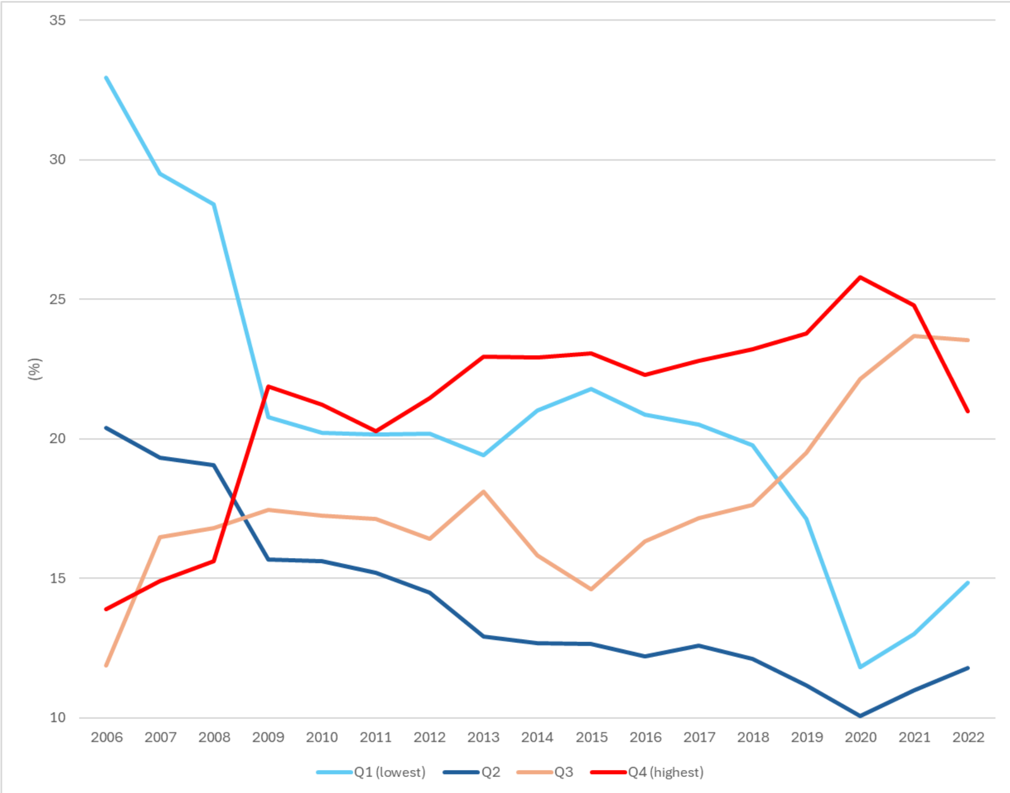
Note: In the figure above, every firm belongs to one and only one quartile for all years.
Source: OECD MAGIC database for steel firms.

This pattern also holds when the sample is split between OECD member and partner economies and quartiles are re-computed for each sub-sample (Figure 11, Figure 12). In OECD countries, the most subsidised firms (Q3 and Q4) tended to lose less market share than their less subsidised peers, while in partner economies the most subsidised firms generally gained market share. The effect was particularly pronounced in 2009: in OECD Member countries, the less subsidised firms lost substantially more market share than more subsidised firms, whereas in partner economies, the most subsidised firms captured the largest gains while their less subsidised competitors both in OECD Member countries and in partner economies lost ground (Figure 11, Figure 12).

Figure 6 builds on the same dataset as Figure 5, using the initial absolute market shares of each subsidy-intensity quartile in 2006 to show how their nominal market shares have evolved over time, i.e., cumulating the above yearly changes.

Figure 6. The most subsidised firms over the study period gained market share at the expense of less subsidised ones

Cumulative changes of market share per (fixed over time) subsidy intensity quartile



Note: In the figure above, every firm belongs to one and only one quartile for all years.
Source: OECD MAGIC database for steel firms.

Firms in the two highest subsidy-intensity quartiles - those receiving the largest average subsidies over the study period - gained market share over time, at the expense of firms in the lower quartiles, which generally lost market share.

2.2. Higher subsidy intensities erode market function

2.2.1. Firms' performance and market share

In a competitive market, improvements in firm-level efficiency - holding other factors constant - should be expected to translate into market share gains. However, subsidies may weaken this link between performance and market outcomes. This section illustrates this hypothesis by looking at correlations.

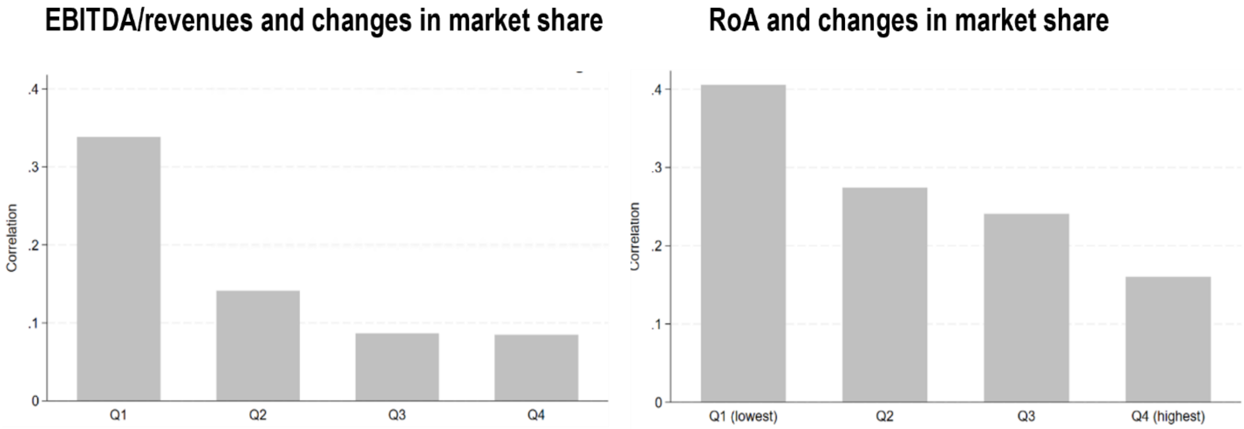
Figure 7 shows the correlation between firms' profitability - measured by EBITDA-to-revenue and return on assets (RoA) - and changes in market share, across subsidy-intensity quartiles from Q1 (least subsidised) to Q4 (most subsidised, relative to asset size).

Although positive across all quartiles, the correlation weakens as subsidy intensity rises. For instance, the link between EBITDA-to-revenue and market share gains is about 33% in Q1, but falls by more than half in Q2 and declines further to below 10% in Q3 and Q4.

The same pattern emerges when alternative performance metrics are used. Substituting EBITDA-to-revenue with net profit margins, or RoA with RoE, produces results consistent with Figure 7: the correlation between performance and market share gains weakens as subsidy intensity rises.

Figure 7. Higher subsidisation weakens the correlation between financial performance and market share gains

Correlations, between firms' financial performance and changes in market share, by subsidy intensity quartile



Source: OECD MAGIC database for steel firms.

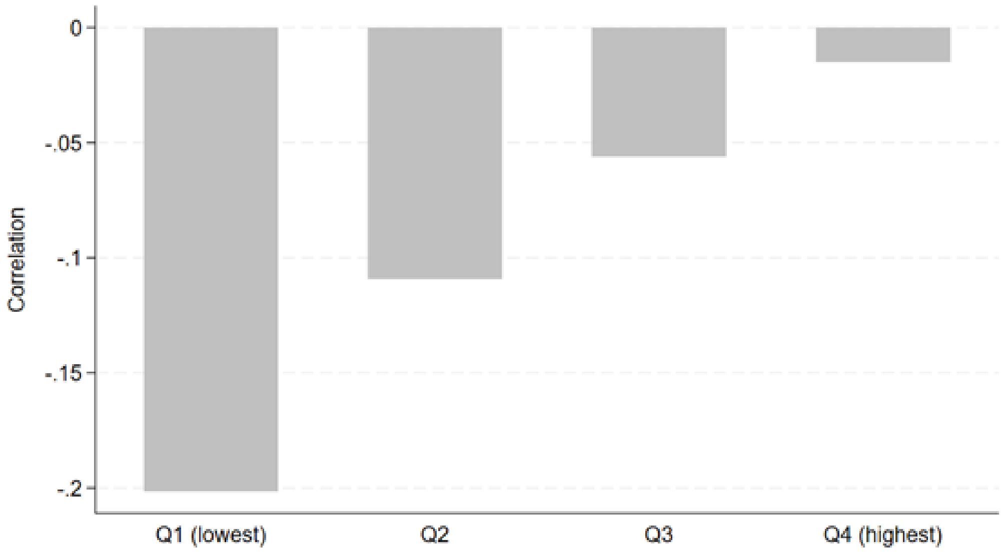
EBITDA includes all fixed operating costs, such as administrative salaries, rents, marketing and sales expenses, non-production utilities, and R&D, as part of operating expenses (OPEX). It excludes interest, taxes, depreciation, and amortisation. As such, EBITDA reflects the core **overall** operational profitability of a steel firm - and is more encompassing than its ability to transform inputs used for its output efficiently.

It is thus useful to complement the EBITDA-to-revenue analysis by focusing on the variable costs of production. This is captured by the cost of goods sold (COGS), which excludes fixed costs. The COGS-to-revenue ratio is a standard operating efficiency metric, with lower values indicating greater production efficiency.

Figure 8 shows a negative correlation between the COGS-to-revenue ratio and market share gains, consistent with economic theory. However, this relationship weakens as subsidy intensity rises and almost disappears for firms in the highest quartile (Q4).

Figure 8. Higher subsidy intensity blurs the correlation between firms' cost efficiency ratio and market share gains

Correlations between COGS/revenue and changes in firms' market share, by subsidy intensity quartile



Source: OECD MAGIC database for steel firms.

To conclude, the correlation patterns above support the hypothesis that subsidies weaken the link between firm performance and market share. This applies both to efficiency - measured by resource use (COGS) and financial structure (debt) - and to profitability.

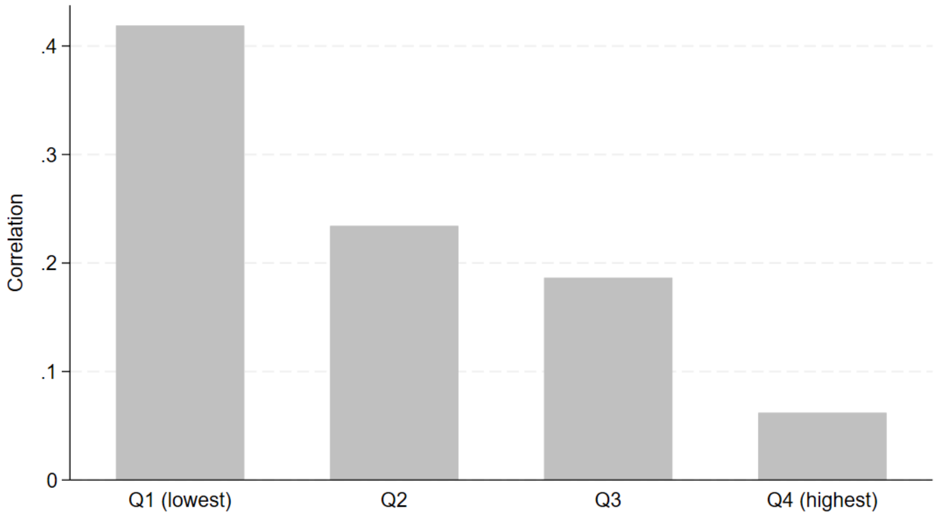
2.2.2. Firms' capacity utilisation and market share

Capacity utilisation is another factor closely linked to firms' market share. A higher use of productive assets typically signals stronger demand and is therefore associated with market-share gains.

Figure 9 confirms this relationship. Yet as the subsidy intensity rises, the correlation between utilisation and market-share gains weakens, suggesting that subsidies dampen the economic link between operational performance - measured here by crude-steelmaking utilisation - and market share.

Figure 9. Higher subsidy intensity blurs the link between capacity utilisation and market share gains

Correlations between changes in capacity utilisation rates (production/capacity) and changes in firms' market share, by subsidy intensity quartile



Source: OECD MAGIC database for steel firms.

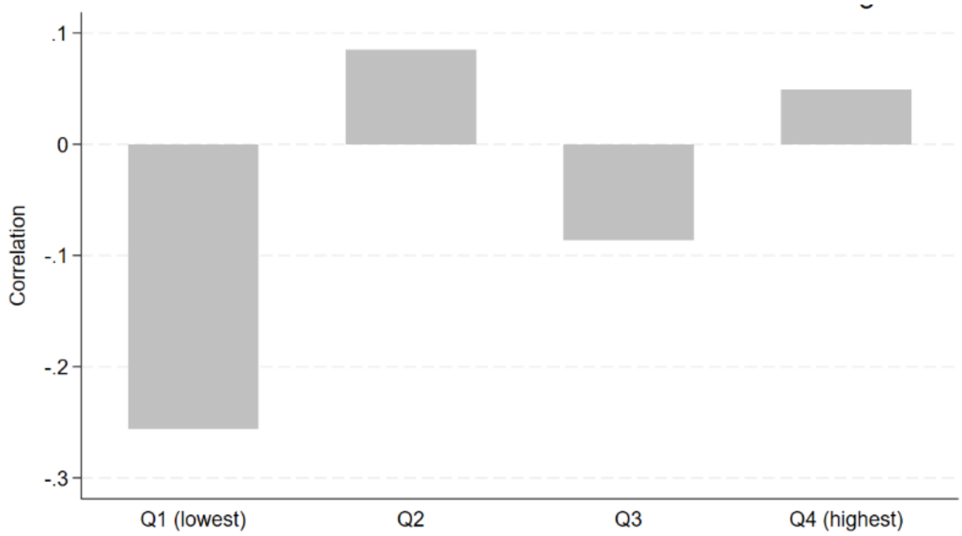
2.2.3. Firms' indebtedness and market share

A high debt-to-asset ratio signals financial fragility and is usually associated with weaker market performance and market-share losses.

Figure 10 shows that this negative correlation is strongest among firms in the lowest subsidy-intensity quartile (Q1). By contrast, it is less pronounced in higher-subsidy quartiles, suggesting that subsidies weaken the link between financial health indicators - such as the debt-to-asset ratio - and market outcomes, even from moderate levels (Q2) onwards (Q3 and Q4).

Figure 10. Higher subsidy intensity is associated with a blurring of the expected correlation between firms' indebtedness and its market share gains

Correlations between changes in the debt-to-asset ratio and changes in firms' market share, by subsidy intensity quartile



Source: OECD MAGIC database for steel firms.

To conclude, higher subsidies appear to blur the expected correlations between market share and its determinants - efficiency, profitability, financial position, and capacity utilisation. In some cases, these correlations disappear altogether for firms in the highest subsidy-intensity quartiles.

These results also hold when OECD Member countries and partner economies are analysed separately, with firms ranked by their relative level of subsidisation within each sub-sample, which is done in the next section.

2.3. The level of subsidisation explains that dynamics differ between OECD Member countries and partner economies

Strong structural differences in market functioning and subsidisation exist between OECD Member countries and partner economies. For example, previous OECD studies found that a typical Chinese steel firm receives 5 times more subsidies in the form of grants and below-market borrowings relative to its revenues than a steel firm located in another partner economy, and 10 times more than a steel firm located in an OECD Member country (Mercier and Giua, 2023[5]), and that the provision of grants and below market borrowing correlates with capacity increases in China and partner economies whereas no statistically significant link was found between the two in OECD Member countries (OECD, 2025[7]). Furthermore, the firm-level characteristics of firms attracting subsidies differed in partner economies compared to OECD Member countries (OECD, 2025[7]).

This section disaggregates the data accordingly. It reproduces the main figures above separately for each subgroup (OECD Member countries, on one side, and partner economies, on the other side) in order to:

- assess if the results hold when quartiles are re-computed with respect to peers (for example firms in Q4 of OECD Member countries may be less subsidised than firms in Q1 of partner economies); and
- enable clearer comparison between the two groups in terms of earning versus losing market shares.

Quantiles in this section (e.g. quartiles in Figure 11, or 50% quantiles used in the next figures) are computed separately within each subsample: OECD Member countries, on one side, and partner economies, on the other. This ensures an equal number of observations in each subgroup. Consequently, the subsidy intensity associated with a given quantile differs significantly between OECD Member countries and partner economies, since the latter display higher subsidy levels on average across the distribution. Recomputing quantiles within each subsample therefore highlights the relative effects of subsidies domestically (more versus less subsidised firms within a group) rather than globally, as done previously, and also allows to draw finer distinctions on the behaviour (and market-winning capacity) of the most vs the least subsidised firms within each sample.

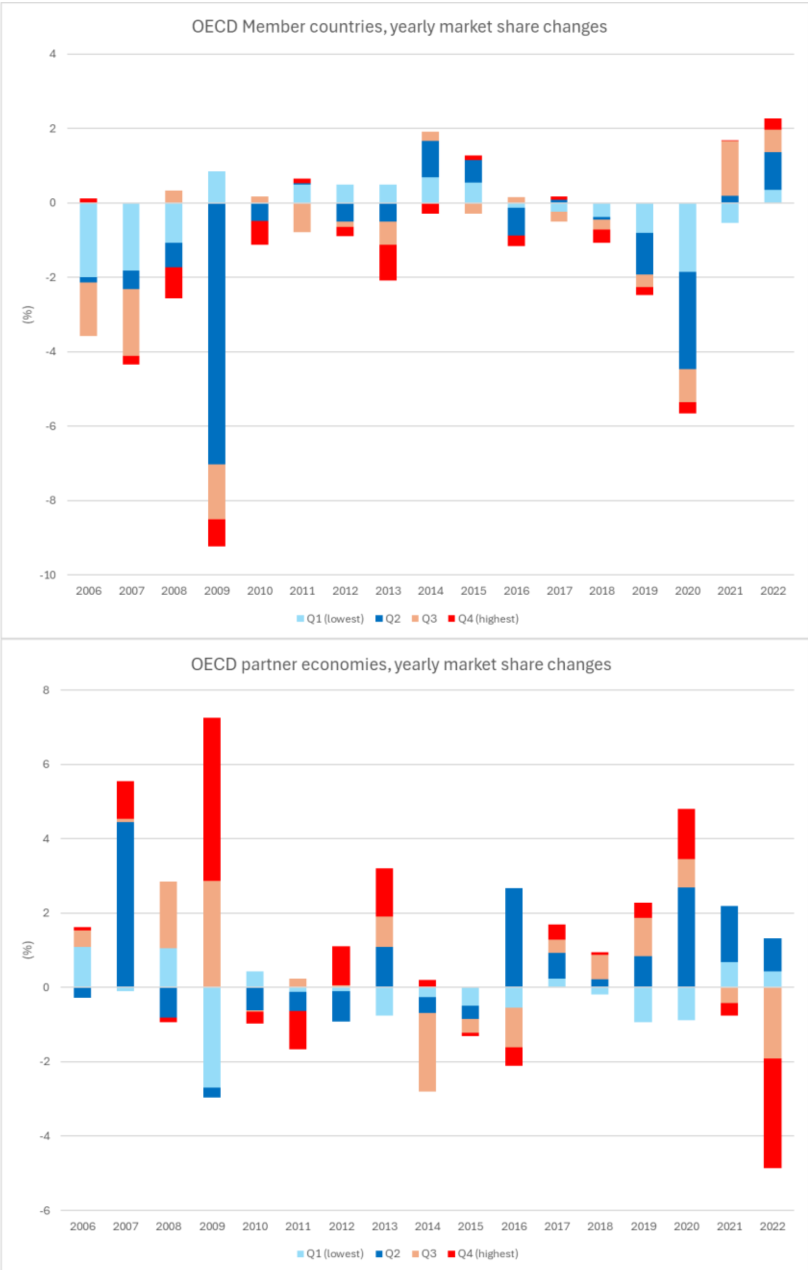
The top graph of Figure 11 shows that in OECD Member countries, firms tended to lose market share through most of the study period, with the least subsidised firms (those in Q1) losing the most - particularly in 2009 and 2020, but the most subsidised firms (in Q4) still losing market shares.

By contrast, in partner economies (bottom graph of Figure 11), the most subsidised firms gained the most market share in 2009. Across the full study period, the least subsidised firms in partner economies (Q1) are the only group to have experienced overall market-share losses (Figure 12). These losses occurred almost continuously, even in years when all other, more subsidised quartiles were gaining share (e.g. 2013, 2018, 2020; Figure 11, bottom graph).

A potential trend reversal emerges in 2021 and 2022: (i) firms in OECD Member countries gained absolute market share; (ii) the most subsidised quartiles in partner economies lost share; and (iii) less subsidised partner-economy firms gained share. This could be related to the effect of the COVID crisis and extensive support measures through loans, guarantees, tax relief and grants to firm in general while economic activity was decreasing, and should probably not be interpreted as a reversal in trend.

Figure 11. Most subsidised firms tend to have better yearly gains than least subsidised firms (with exceptions in 2021, 2022)

Yearly changes of market share per (fixed over time) subsidy intensity quartile

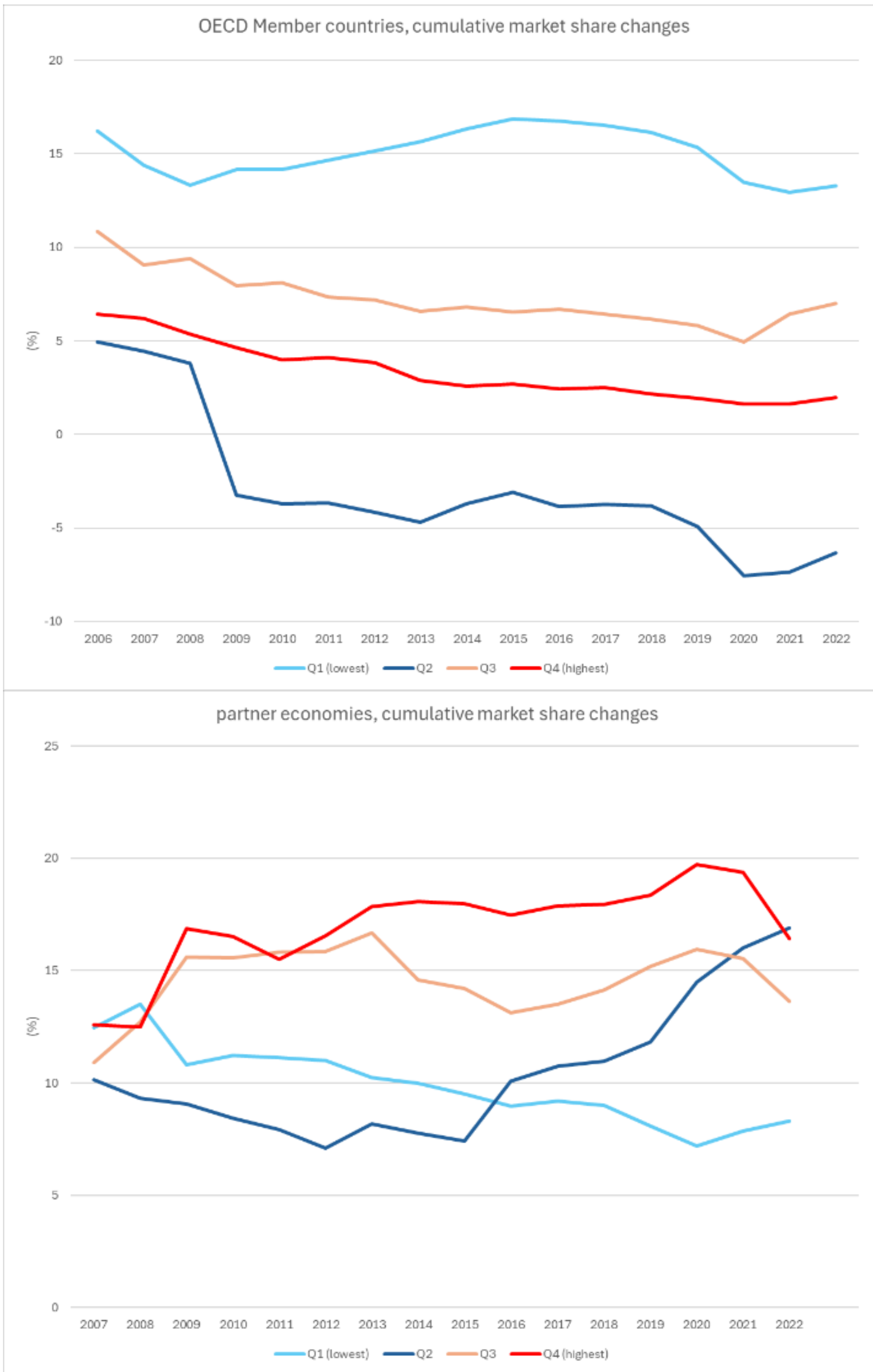


Note: In both graphs, each firm is assigned to a single quartile for all years. Quartiles are computed separately within each subsample, so each quartile contains the same number of observations. As a result, the subsidy intensity represented by a given quartile differs between OECD Member countries and partner economies, since firms in partner economies generally exhibit higher subsidy intensities.
Source: OECD MAGIC database for steel firms.

Figure 12 builds on the same data as Figure 11 adding the initial 2006 market share of each subsidy-intensity quartile to show how nominal market share evolved over time.

Figure 12. Most subsidised firms over the study period earned market share at the expense of less subsidised ones, or lost less market share than less subsidised ones

Cumulative changes of market share per (fixed over time) subsidy intensity quartile

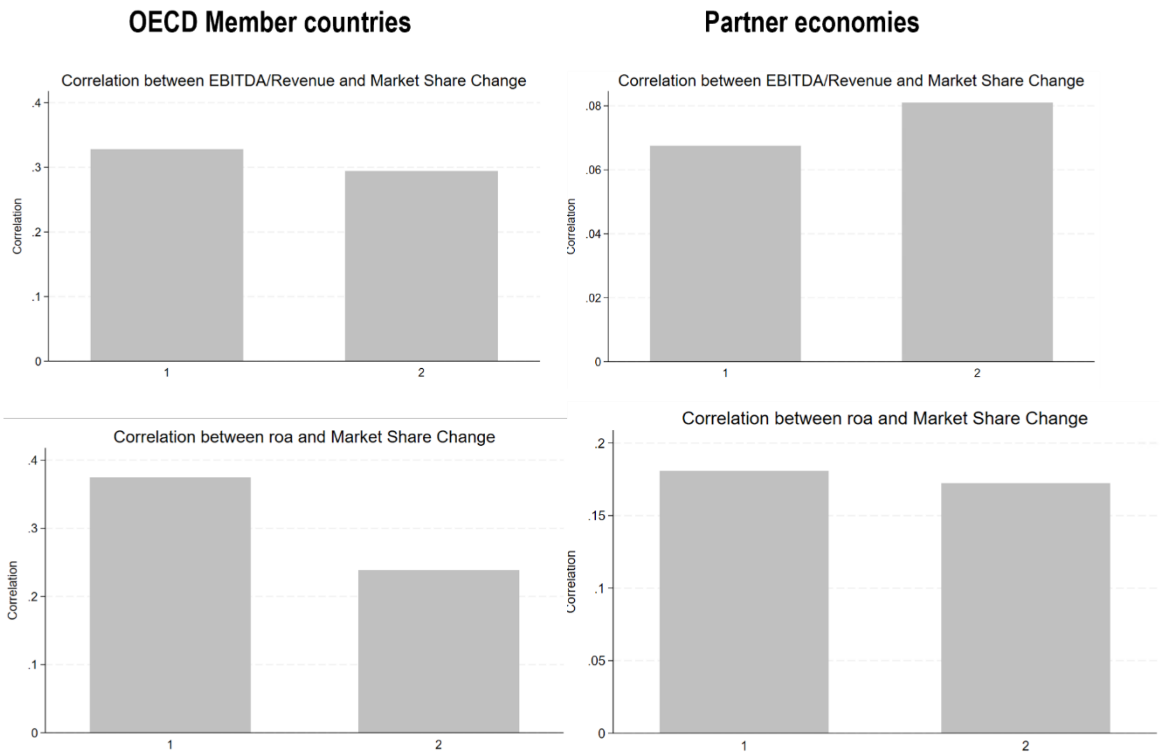


Note: In both graphs, each firm is assigned to a single half (lower or upper 50%) for all years. These 50% quantiles, based on subsidy intensity, are computed separately within each subsample, ensuring equal numbers of observations in each half. As a result, the subsidy intensity associated with a given half differs between OECD Member countries and partner economies, since firms in partner economies generally exhibit higher subsidy levels.
Source: OECD MAGIC database for steel firms.

Figure 13 below indicates that the weakening of the correlation between performance indicators and market share is less pronounced in partner economies than in OECD Member countries. However, this mainly reflects the much lower initial correlations in partner economies (e.g. 6.2% between EBITDA-to-revenue and market share, versus 32% in OECD Member countries).

Figure 13. Higher subsidy intensity is associated with a decrease in the size of the positive correlation between firms' performance and its market share gains

Correlations, between a firms' performance and changes in market share, by subsidy intensity quartile

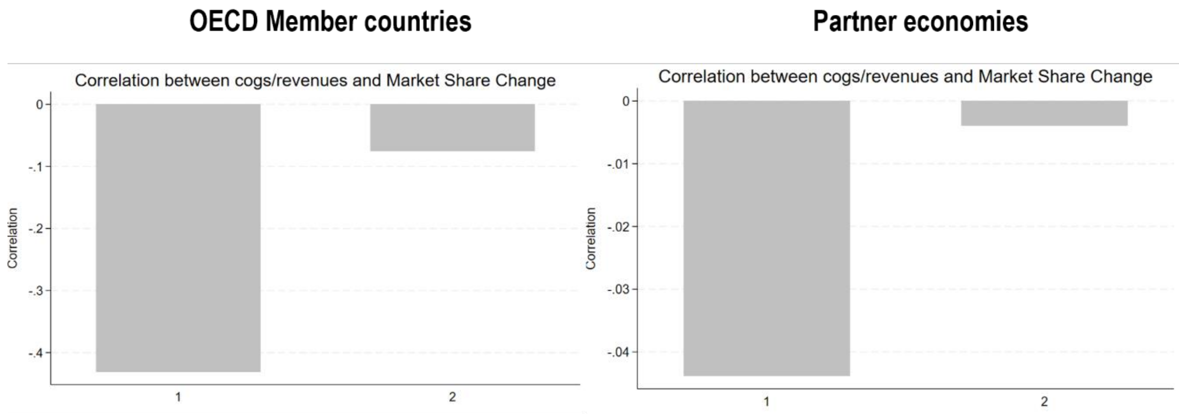


Source: OECD MAGIC database for steel firms: 49 firms, 806 observations.

For the operating efficiency ratio (COGS-to-revenue), where lower values indicate greater production efficiency, the correlation with market-share gains erodes similarly in both OECD Member countries and partner economies. Partner economies, however, start from a much lower initial correlation and still experience the steepest relative decline as subsidy intensity rises, with the correlation falling to nearly zero (Figure 14).

Figure 14. Higher subsidy intensity blurs the correlation between firms' cost efficiency ratio and market share gains

Correlations, between COGS/revenue and changes in firms' market share, by subsidy intensity quartile



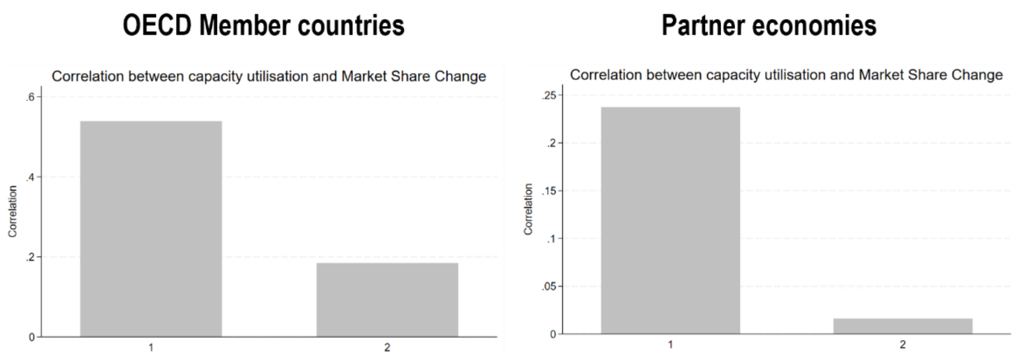
Source: OECD MAGIC database for steel firms: 49 firms, 806 observations.

Those correlation patterns confirm that *subsidies blur the relationship between market share and its usual determinants* - efficiency, capacity utilisation, and profitability - in both OECD Member countries and partner economies, and that correlations are multiples lower to start with in partner economies, probably due to higher subsidisation levels of the lowest quantiles.

Capacity utilisation rates are another factor typically linked to firms' market share. Figure 15 confirms this relationship in both OECD Member countries and partner economies, though correlations are weaker in the latter. As subsidy intensity rises, the link between capacity utilisation and market-share gains weakens, suggesting once again that subsidies dampen the natural connection between firm performance - specifically, the effective use of crude steelmaking capacity - and market outcomes.

Figure 15. Higher subsidy intensity is associated with a decrease in the size of the positive correlation between firms' capacity utilisation and market share gains

Correlations, between changes in capacity utilisation rates (production/capacity) and changes in firms' market share, by subsidy intensity quartile



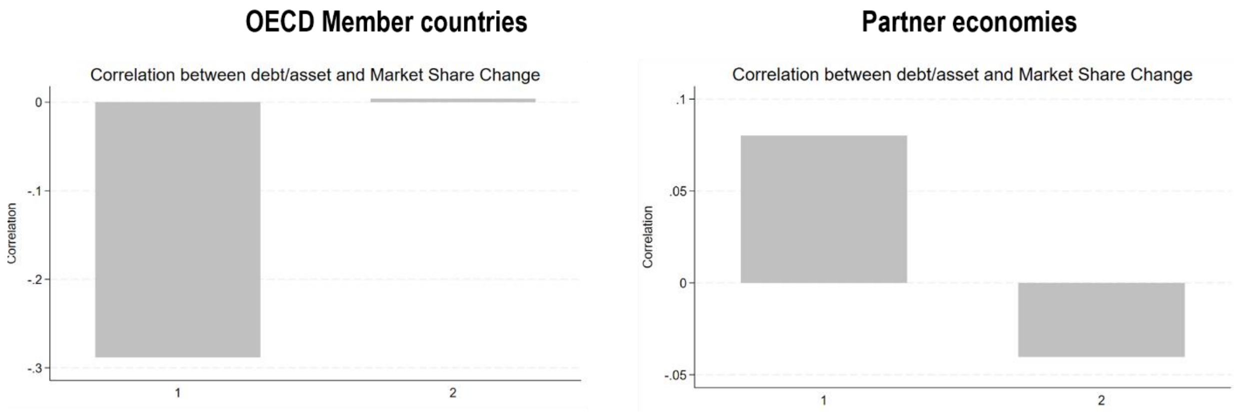
Source: OECD MAGIC database.

High indebtedness, measured relative to total assets, generally signals a more fragile financial position. This is typically associated with weaker market performance and, by extension, losses in market share.

For partner economies, Figure 16 shows very low - and likely insignificant - correlations (around 7%), making interpretation difficult. By contrast, in OECD Member countries, the expected negative correlation is evident among firms with low subsidy intensity (Q1), at a significant -30%, but disappears for firms with higher subsidy intensity (Q2). This pattern supports the hypothesis that subsidies attenuate the influence of financial health on market outcomes.

Figure 16. Higher subsidy intensity is associated with a decrease in the size of the negative correlation between firms' indebtedness and market share gains

Correlations between changes in the debt-to-asset ratio and changes in firms' market share, by subsidy intensity quartile



Source: OECD MAGIC database.

3 Econometric estimations of the impact on subsidies on market share

This section applies Ordinary Least Squares (OLS) regressions to examine the association between market-share gains and subsidy intensity, controlling for firm-level characteristics. Although a separate panel analysis on the OECD Member countries and partner economies subsamples would have been useful in theory, due to the limited number of observations it results in all the variables (including subsidy intensity) losing statistical significance.[3] Nevertheless, although the impact of subsidies on market share works for both OECD Member countries and partner economies, that impact is much higher for partner economies given the much higher subsidy intensity in those economies.

3.1. Econometric specifications

Since subsidies to other firms can affect market share through competitive dynamics, the key variable is not the absolute subsidy intensity but its deviation from the sample average. Specifically, for each firm j in year t, the subsidy-intensity deviation is defined as:

$$\mathrm{dev}\left(\frac{\mathrm{subsidy}}{\mathrm{asset}}\right)_{j,t} := \frac{\mathrm{subsidy_{j,t}}}{\mathrm{asset_{j,t}}} - \frac{1}{n_t}.\sum_{all\ firms\ i} \frac{\mathrm{subsidy_{j,t}}}{\mathrm{asset_{j,t}}}$$

with n_t representing the number of firms in the sample in year t.

OLS regressions are estimated in a panel-data framework, with all explanatory variables lagged one year. This lag structure reduces potential reverse causality between the subsidy received and market-share gains. Consequently, the coefficients capture delayed rather than contemporaneous effects, reflecting the time it takes for subsidies to influence market performance - for instance, through investment responses or sustained price undercutting.

The main regression equation is specified as described in Equation 1 below:

$$\mathrm{d.}(MarketShare)_{j,t} = \alpha\,\mathrm{L.dev}\left(\frac{\mathrm{grants}}{\mathrm{asset}}\right)_{j,t} + \beta\,\mathrm{L.dev}\left(\frac{\mathrm{BMB}}{\mathrm{asset}}\right)_{j,t} + \gamma.\,\mathrm{L.dev}\left(\frac{\mathrm{income\ tax\ concessions}}{\mathrm{asset}}\right)_{j,t} + \sum_{k} \gamma_k\,\mathrm{L.d.X}_{k,j,t} + \mu_t.\,year_t + \varepsilon_{j,t}$$

where d. is the first-difference operator, L. the lag operator, $\mathrm{X_k}$ the control variables (debt-to-asset ratio, EBITDA-to-revenue, capacity utilisation, etc.) and $year_t$ year-fixed effects.

An alternative specification distinguishes between a firm's own subsidies and the average subsidies received by its competitors, allowing a decomposition of competitive spillover effects. The specification can be expressed as in Equation 2 below:

$$
\begin{aligned}
\mathrm{d.}(MarketShare)_{j,t} = {} & \alpha\,\mathrm{L.}\left(\frac{\text{grants}}{\text{asset}}\right)_{j,t} + \alpha'\mathrm{L.}\left(\frac{1}{(n_t-1)}\cdot\sum_{all\ firms\ i\neq j}\left(\frac{\text{grants}}{\text{asset}}\right)_{j,t}\right) \\
& + \beta\,\mathrm{L.}\left(\frac{\text{BMB}}{\text{asset}}\right)_{j,t} + \beta'\mathrm{L.}\left(\frac{1}{(n_t-1)}\cdot\sum_{all\ firms\ i\neq j}\left(\frac{\text{BMB}}{\text{asset}}\right)_{j,t}\right) + \gamma.\,\mathrm{L.}\left(\frac{\text{tax rebates}}{\text{asset}}\right)_{j,t} \\
& + \gamma'\mathrm{L.}\left(\frac{1}{(n_t-1)}\cdot\sum_{all\ firms\ i\neq j}\left(\frac{\text{tax rebates}}{\text{asset}}\right)_{j,t}\right) + \sum_{k}\gamma_k\,\mathrm{L.d.X}_{k,j,t} + \mu_t.\,year_t + \varepsilon_{j,t}
\end{aligned}
$$

Results from estimations (and slight variations) of the two equations above are presented in Section 3.2 below.

3.2. Results

Results for the first and second specifications are shown in Table 1 and Table 2, respectively. Because the models are in first-difference, including a constant amounts to imposing an exogenous linear trend in market share, yet estimates without a constant are similar (hence not reported here). Year-fixed effects are also insignificant in both specifications.

Table 1. Results from OLS regression for market share gains (using deviations from the mean, i.e. Equation 1)

d.(Market share)	1	2	3	4	5	6	7	8	9
L.dev(grants/asset)	0.152035	0.1987706*							
L.dev(BMB/asset)	0.0338735**		0.0398121**				0.0514229*	0.061371**	
L.dev((grants+BMB)/asset)				0.0400099**	0.0398185**	0.0328214*			0.0532892*
L.dev(tax rebate/asset)							0.1771462*	0.1231429	0.1591257*
L.d.(debt/asset)	0.015385***	0.013040***	0.012515***	0.012632***	0.015188***	0.014306***	0.017750***	0.017477***	0.012753***
L.d.(EBITDA/revenue)	0.0074402*	0.0084722**	0.0083026**	0.0083601**	0.0072182*	0.0081101**	0.0055118	0.0079189*	0.0095199**
L.d.(capacity utilisation)	0.0011028	0.0010448	0.0012476	0.0012551	0.0012478		0.0019901		0.001833
dev(capacity utilisation)						0.004258***		0.005963***	
Constant	yes	yes	yes	yes	yes	yes	yes	yes	yes
Fixed year effect	Yes but not significant	no	no	no	yes	yes	Yes but only 2009 significant	Yes but only 2009 significant	no
R2	6%	3%	3%	3%	6%	6%	11%	14%	5%
Nbr of observations	503	503	526	503	503	525	390	395	388

Note:* indicates a statistically significant result at the 10% confidence level, ** at the 5% confidence level and *** at the 1% confidence level. Results are very close whether we use the average intensity ratio of all firms or of competitor firms. The means indicated in the definition of the operator dev() represent the means over all the firms in the sample, but using instead the mean over all *competing* firms in the sample gives almost similar results.

The results indicate that subsidy intensity - measured by grants-to-assets, BMB-to-assets, or their sum - is significantly and positively associated with market-share gains. This finding is consistent with a recent OECD multi-sector study (OECD, 2025[4]), which identified the same effects using three distinct econometric techniques. However, when grants and BMB are included jointly, the coefficient on grants becomes insignificant, likely due to multicollinearity. Among the controls, EBITDA-to-revenue and debt-to-assets are both significant and positively correlated with market-share gains, as expected. Capacity utilisation is significant only when measured as a deviation from peers' average utilisation, which filters out industry-wide cyclical effects; in this specification, it is included contemporaneously.

Corporate income tax concessions are also considered as a form of subsidy. Estimates are available in the OECD MAGIC database for a subset of firms, primarily based on corporate disclosures and, in some cases, adjusted or estimated by the OECD. However, including tax concessions creates strong multicollinearity with EBITDA-to-revenue, since both capture aspects of profitability. The results are reported in columns 7 and 8 of Table 1.

Table 2 reports results from a specification that separates a firm's own subsidy intensity from that of its competitors, by instrument type. Although some coefficients fall just short of the 10% significance threshold, their signs align with economic expectations: a firm's market share tends to rise with its own subsidies (grants, BMB, or tax concessions) and fall with the average subsidies received by rivals. These patterns hold in both the full model (Equation 1) and the alternative specifications (Equations 2 to 7). Overall, the results point to negative competitive spillovers from subsidisation.

Table 2. Results from OLS regression for market share gains (estimating separately effects from others' subsidies from own, i.e. Equation 2)

d.(Market share)	1	2	3	4	5	6	7
L.(grants/asset)	0. 1998427	0.1915618*					
L.mean(grant/asset)	-0.6546059	-0.2973983					
L.(BMB/asset)	0 .0295708		0.0366368*				0.0374646
L.mean(BMB/asset)	-0.1967383*		-0.1004729				-0.1802668**
L.((grants+BMB)/asset)				0.0367463*		0.042375	
L.(mean(grants+BMB)/asset)				-0.1125339		-0.195173**	
L.(tax rebate/ asset)	0.1507192				0.1648242*	0.1581163*	0.1594*
L.mean(tax rebate/ asset)	-0.144232				0.0934393	-0.219291	-0.2673481
L.d.(debt/asset)	0.0133517***	0.0129202 ***	0.0129191 ***	0.0130245 ***	0.0125226 ***	0.0132727 ***	0.0131719 ***
L.d.(EBITDA/revenue)	0.0088367**	0.0085865**	0.0077363**	0.0077907 **	0.0090033*	0.0084585**	0.0083783**
L.d.(capacity utilisation)	0.0005943	0.0010281	0.0009316	0.0008451	0.0016284	0.0009144	0.0010672
Constant	yes	Yes but not significant	yes	yes	yes	yes	yes
R2	6%	3%	3%	4%	6%	6%	5%
Nbre observations	503	503	526	503	503	525	390

Note:* indicates a statistically significant result at the 10% confidence level, ** at the 5% confidence level and *** at the 1% confidence level. The means indicated represent the means over all competing firms in the sample.

4 Discussion of the findings and policy considerations

This report shows that subsidies - whether in the form of grants or below-market borrowings (BMB) - weaken the normal correlation between (global) market shares and the market drivers that determine these shares. The report shows that firms that receive subsidies gain market share at the expense of less subsidised or unsubsidised competitors.

A recent OECD study finds that subsidies have no significant positive effect - or even a negative effect - on productivity (OECD, 2025[4]). The market-share gains observed here are therefore likely to result from recipients' ability to lower prices and undercut competitors, or from discouraging rival firms from investing and expanding capacity.

An additional explanation is that firms subsidised through grants, BMB, or tax concessions may also benefit from demand-side support, such as preferential access to public procurement, which can shift market share in their favour. This procurement aspects could be relevant for further analysis.

The impact of subsidies can therefore operate through two channels: (i) direct diversion of market share towards subsidised firms via policy support, and (ii) reduced production costs that enable price undercutting in competitive or less-protected markets.

Figure 17 illustrates how other common subsidy instruments - beyond the grants, BMB, and tax concessions analysed in this report - can contribute to these effects. The name of subsidy instruments covered in MAGIC appear in blue, while instruments not covered in MAGIC appear in black text. Market-reservation effects are described in violet, and price-undercutting enablers in red. Because different subsidy instruments are often correlated, their impacts may be partly captured by the estimated effects of the three instruments covered in MAGIC. Accordingly, grants, BMB, and tax concessions can be seen as proxies for broader subsidisation patterns involving multiple instruments.

The weakening correlation between market-share outcomes and performance indicators - such as profitability, capacity utilisation, and leverage - among more subsidised steel producers indicates that **subsidies distort well-functioning market outcomes**. By dampening the market signals that would normally reward more efficient firms - those that use resources effectively, add greater value, or operate with lower debt - subsidies risk reducing overall economic efficiency and, ultimately, aggregate welfare.


Subsidies to
input providers
lower costs of
inputs
Exports
Barriers to
raw material
exports
Tax exemptions
for exports,
export subsidies,
etc.:
• lower export prices
Exports

Based on these findings, the report proposes the following policy recommendations, in line with wider OECD work on subsidies and industrial policies:

- **Strengthen transparency and discipline in subsidy allocation:** Subsidies - whether provided as grants, BMB, or corporate tax concessions - undermine the normal relationship between market share and firm performance. This decoupling allows subsidised firms to gain share at the expense of more efficient, unsubsidised competitors, distorting competition and reducing overall welfare. Greater transparency in subsidy allocation and closer alignment with clearly defined policy objectives can help limit these distortions. This recommendation has been made forcefully in the result of the joint-work of the International Monetary Fund (IMF), the World Bank (WB), the World Trade Organisation (WTO) and the OECD (IMF et al., 2022[3]).
- **Avoid price-distorting forms of support:** Since subsidies have shown no evidence of boosting productivity (OECD, 2025[4]), market-share increases among supported firms are more likely driven by price undercutting or by discouraging competitor investment, with possible additional effects from direct market diversion. To preserve incentives for efficiency and innovation, governments should avoid support measures that artificially reduce marginal costs, as these distort price formation - particularly in internationally traded sectors.
- Monitoring of subsidies in the steel sector would benefit from **parallel monitoring of steel market shares of subsidised and non-subsidised firms**. This could be done at the level of International Trade Administration, with a concerted effort in the Steel Committee or the GFSEC, and could include a split-off between domestic and non-domestic market shares in case the dynamics at play differ.
- **Assess cumulative effects of overlapping subsidies:** Firms subsidised through grants, BMB, or tax concessions may also receive demand-side support, such as preferential access to public procurement. This bundling of support can amplify market-share shifts. Policymakers should therefore systematically evaluate how different subsidy instruments interact and their joint impact on market functioning, such as steel (Criscuolo and Lalanne, 2022[9]).
- **If subsidies are provided, ensure that they are targeted to address well-defined market failures:** To minimise distortions, subsidies should be designed to address clearly identified market failures - such as externalities, public goods, or coordination failures - rather than to prop up individual firms or sectors. Broad-based, untargeted, or firm-specific subsidies pose a higher risk of misallocation and efficiency losses. Targeted industrial strategies (e.g. mission-oriented or technology-focused) can steer technological change and support growth, notably in addressing societal challenges such as climate change. However, they require careful governance to avoid capture, ensure inclusiveness, specify objectives clearly, plan regular evaluation, and include exit options (Criscuolo and Lalanne, 2022[9]).
- **Safeguard competitive neutrality:** The weakening link between firm performance (profitability, capacity utilisation, indebtedness) and market-share outcomes in more heavily subsidised firms reflects reduced competitive neutrality. Subsidies should not shield firms from the competitive pressures that reallocate resources toward more productive and financially sound producers. Soft budget constraints that let firms expand in upturns while avoiding adjustment in downturns must be avoided. The countercyclical use of BMB in some partner economies (OECD, 2025[7]) is especially concerning, as it may block resource reallocation toward more

efficient producers. Aligning subsidy policy with principles of competitive neutrality is essential to safeguard long-term efficiency and aggregate welfare. Governments should institutionalise the regular evaluation of targeted policies (Criscuolo and Lalanne, 2022[9]).

- **Efforts to reduce subsidies provided to the steel sector should be pursued in a coordinated manner.** The findings of this report indicate that an uncoordinated withdrawal of subsidies - for instance, limited to OECD Member countries - could further distort market shares, as the analysis shows that firms' market positions are influenced by the relative scale of subsidies received by their competitors. A rapid phase-out of support measures in jurisdictions that are not among the most subsidy-intensive could therefore inadvertently strengthen the market share of more heavily subsidised firms, even without an increase in their own support levels.

Tr		discriminatory government procurement		dual pricing					
	Including advantages conferred through state enterprises			Provision of below-cost electricity by a state-owned utility			Below-market loan by a state-owned bank		

Source: (OECD, 2023[11])

Annex B. The OECD MAGIC database

The OECD MAnufacturing Groups and Industrial Corporations (MAGIC) database is a confidential firm-level database combining financial and economic data and estimates of government support at the level of each industrial group covered over the period 2005-22. It aims to help improve understanding of the scope and scale of government support in manufacturing and to enable analysis of how this support affects firms' decisions and markets.

The OECD MAGIC database does not aim to cover all manufacturing sectors but instead concentrates on sectors producing either durable goods (e.g. capital goods) or industrial raw materials (e.g. aluminium, steel, and chemicals) (Table A B.1). Preference is given to products destined for other businesses (B2B), with the notable exception of automobiles, which are purchased by both businesses and households. This implies that the database does not cover manufacturing industries such as food products, beverages, tobacco products, textiles, wearing apparel, leather and related products, and domestic appliances. Services are also not covered unless they are provided attached to or embodied in manufactured products.

Table A B.1. Sector coverage of the OECD MAGIC database (version 1.0)

Short sector name	Full sector name	ISIC Rev.4 correspondence
AERO	Aerospace & defence	3 030
ALUM	Aluminium	2 420
AUTO	Automobile	2 910
CEMT	Cement & other building materials	2 394
CHEM	Chemicals	2 011, 2 013, 2 02X, 2 03X
FERT	Fertilisers	2 012
GLAS	Glass, ceramics & refractories	2 31X, 2 391, 2 392, 2 393
SEMI	Semiconductors	2 610
SHIP	Shipbuilding	3 011
SOLA	Solar photovoltaic cells & modules	2 710
STEE	**Steel**	**2 410**
TELC	Telecommunications network equipment	2 630
TRAN	Rolling stock & signalling	3 020, 2 790
WIND	Wind turbines	2 811

Version 1.0 of the OECD MAGIC database covers 482 consolidated industrial groups – including both publicly listed and non-listed companies – over the period 2005-22, where applicable and depending on data availability. These firms were selected because they are the largest ones globally by sales revenue, output, or capacity in each of the 14 industrial sectors listed in Table A B.1. For steel, this includes 47 of the largest steel firms worldwide, which collectively accounted for about half of global steelmaking capacity and about 65% of global revenue for the steel sector in 2022.

The firms covered in the OECD MAGIC database are based in several different jurisdictions, with the resulting geographical split depending on these jurisdictions' relative strength in global manufacturing. The home jurisdiction of the companies is taken to be either the location of their headquarters or their main place of business (e.g. measured by tangible assets or employees) in cases where the location of their headquarters corresponds to inexistent or insignificant volumes of manufacturing activity. This helps mitigate the issue of multinationals choosing to locate their headquarters in low-tax jurisdictions. In the case of the steel firms covered here, the sample comprises 13 firms based in the OECD and 34 based outside the OECD. These firms represent respectively 36% and 62% of total steelmaking production capacity for 2022 in OECD Member countries and partner economies respectively.

A consequence of looking at what firms obtain – as opposed to looking at what governments provide – is that the OECD MAGIC database does not distinguish between subsidies based on their policy objectives or design characteristics (e.g. eligibility criteria). Thus, the data do not make a distinction between one-off crisis measures and longstanding industrial policies seeking to affect firm behaviour durably. That said, in showing the persistence of various forms of support over time the database can shed light on the role of government support in shaping the market in a given sector. Moreover, while the MAGIC database does not differentiate between measures targeting R&D and those targeting investment in new industrial capacity, these issues have been explored in related sectoral studies by the OECD, for example in the semiconductor sectors (OECD, 2019[12]) and the rolling-stock (OECD, 2023[13]).

The above caveats notwithstanding, the OECD MAGIC database has the important advantage of capturing all support received by firms, irrespective of which jurisdiction provides it. This avoids selective coverage of policies caused by lack of government transparency and by the existence of numerous subsidy programmes at sub-national levels. It also helps capture the support provided through state enterprises such as state banks (OECD, 2024[8]), which might otherwise fall outside the perimeter of national budget reporting.

As of version 1.0, the OECD MAGIC database covers three types of subsidies, namely government grants, corporate income tax concessions and below-market borrowing:

- Annual values for government grants are obtained directly from primary sources, which are generally corporate disclosures, at times complemented by government sources (all the while avoiding double counting). It has to be noticed that recorded grant amounts do not always represent the actual money flow provided to a company on a given year, as explained below in Box 1.
- Estimates of corporate income tax concessions in the OECD MAGIC database do not refer to the tax revenue foregone by governments (i.e. tax expenditures) but to the tax savings of companies due to particular provisions of the tax code of the jurisdictions in which these companies operate. The estimates are primarily based on corporate disclosures and involve some amount of interpretation, judgment, and estimation by the OECD, such as where only certain subsidiaries of a group are eligible to a preferential rate of corporate income tax. Tax concessions are generally less internationally comparable than government grants owing to international differences in statutory tax rates.
- Below-market borrowings are estimated by the OECD by comparing actual interest rates that firms are charged on their borrowings against hypothetical benchmark interest rates that would normally prevail in the market based on borrowers' financial profile. The resulting differences are multiplied by the amount of debt to arrive at the volume of below-market borrowings for each year. Because below-market borrowings generally involve state banks offering loans at below-market rates, they tend to be much more common in jurisdictions in which the state plays a large role in the financial system.

While the MAGIC database represents a very significant improvement in the availability of data on industrial subsidies, it does not exhaust all possible ways in which governments can subsidise industrial companies. The full spectrum of possible support measures can include, for example, the effect of differential treatment in relation to regulatory measures or support resulting from export restrictions on upstream inputs. OECD estimates of the monetary value of below-market energy inputs exist for several firms and sectors but do not yet have sufficient coverage to warrant inclusion in the MAGIC database (OECD, 2023[14]). Estimates of the value of land acquired or rented by firms at below market prices are, for their part, non-existent due to a lack of sufficient data and methodological difficulties. However, anecdotal evidence does indicate that below-market land may confer significant benefits to certain industrial firms. These examples suggest that there is still considerable room for further improvements in the measurement of industrial subsidies and analysis thereof.

Monetary variables in the OECD MAGIC database are expressed in millions of current USD. They are converted from firms' original reporting currencies into current USD using nominal, period average exchanges rates from the OECD and the IMF for every year covered by the OECD MAGIC database. Since many firms in the database are multinationals operating across several jurisdictions, the choice is made not to correct for inflation given the complications involved in choosing adequate deflators. This implies that changes in subsidy amounts (where not expressed in % of revenue) can reflect policy changes but also differences in inflation rates and exchange rate movements.

While the OECD makes its best efforts to cover the entire period 2005-22, there are several companies in the MAGIC database for which year coverage is shorter than 18 years. In most cases, this results from firms having entered a sector covered after 2005 (late entry), firms having left a sector covered before 2022 (early exit), or firms having merged with or having been acquired by another entity. In fewer cases, this does concern data that are genuinely missing. Overall, time coverage is generally good for the period 2010-22 but can be more limited in certain sectors (e.g. aluminium, shipbuilding, and solar cells and modules) over the period 2005-09.

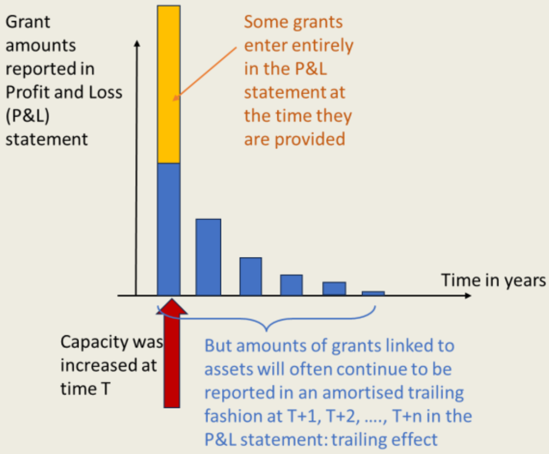
Since firms were selected to cover as much of world steel production as possible (as opposed to be representative of each individual jurisdiction), the data certainly covers large steel players more than the smaller steel firms. It may or may not lead to higher level of government support than those witnessed in a "typical", median steel firm. On one side, subsidies received by the large steel firms included in the sample may over-represent the subsidies received by smaller firms, since the fixed costs of applying for subsidies can be more easily borne by larger firms, and larger firms may be better positioned for lobbying government officials to obtain subsidies. On the other side, non-transparent steel firms for which the annual reports were not accessible through desk research were not included in the sample for obvious reasons, and those may be large recipients of government support. All in all, it is still necessary to exercise caution when assigning the results of the present study to a "typical" firm, especially if the firm is of a small size, as the results are probably not authoritative in covering all steel producing firms in any of the major steel-producing economies represented in the sample.

Box 1. Grants amounts in the Profit and Loss statement

The amounts of grants used in the regressions correspond to the amount of grants entering the Profit and Loss statement of the annual reports of a given steel firm for the given year. This approach was chosen for the OECD MAGIC database in order to ensure comparability across jurisdictions, but it entails a drawback for its economic (as opposed to financial reporting) interpretation. The drawback is that each year, the actual money amount of subsidies provided through cash grants (i.e., the nominal amount of the disbursed grants) does not necessarily correspond with the recorded amount for the grant in the database. This is because many grants are attached to assets and, according to accounting standards, companies will only report their amounts in an amortised fashion over the whole life of the said assets.

Due to this accounting difference, many grant amounts entering the Profit and Loss statement and hence being recorded in the variable "grants" used in this paper for regression analysis correspond to a grant of a much larger magnitude being received by the steel firm, whose total amount is available for purchase or building capacity immediately. Furthermore, they will have a trailing effect in the sense that a grant received in year T will continue providing grant amounts in years T+1, T+2, … T+n, where n is the amortisation period of the associated asset (in years), even if no grants are received in those years T+1, T+2, … T+n, as depicted in Figure 18 below.

Figure 18. In accounting there are two main categories of grants



The situation depicted in Figure 18 only corresponds to a single grant linked to asset (that amortises) and all the grants that directly enter the profit and loss statement at time T. In practice, there is a superposition of many such grants, each either entering the profit and loss statement entirely at a given time, or with its own amortisation schedule (linear or exponential).

Annex C. Estimating below market borrowing

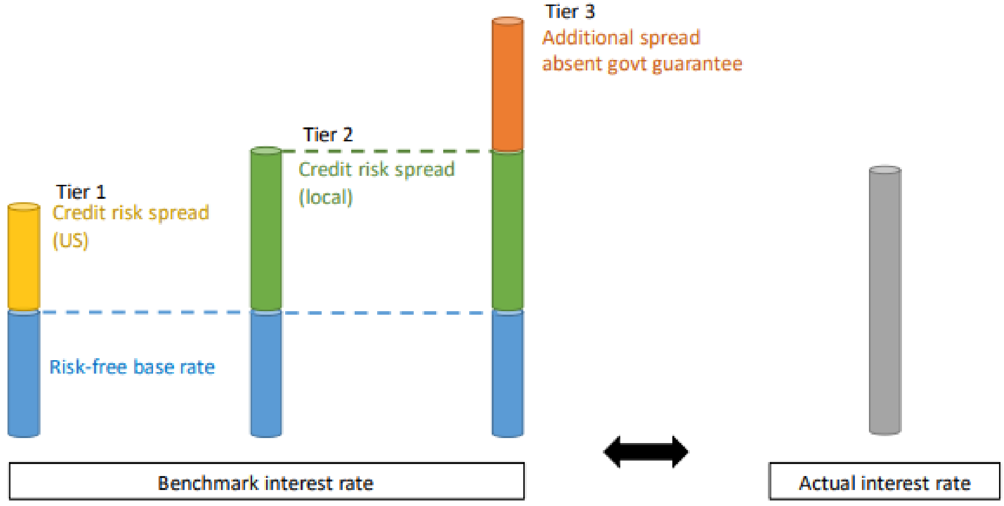
The estimation of below-market borrowings compares the actual interest rates charged to firms against hypothetical benchmark interest rates that could have been charged in a private market, based on the characteristics of the borrower, as laid out in (OECD, 2021[15]).

Since information is not available on every single interest rate charged to all firms covered in the MAGIC database, firms' annual average interest rates are computed by dividing their interest payments in any given year (t) by the average debt outstanding in the same year (t) and the previous year (t-1). Information necessary for performing this calculation is retrieved from the financial statements of each firm, wherever available.

Calculating rates in this way ensures consistency across the sample of firms. The benchmark interest rates are constructed following established finance principles by combining a risk-free base rate and additional spreads that reflect the credit risk of a borrower. Depending on their availability and on each country's practice, risk-free base rates include and are chosen from: common banking reference rates (e.g. the US Secured Overnight Financing Rate, Euro Interbank Offered Rate, Tokyo Interbank Offered Rate, etc.); one-year government bond yields; or other commonly used base rates (one year), such as the base rates published by the People's Bank of China (e.g. the Loan Prime Rate).. As the currency of benchmark rates should ideally match that of the corporate debt being analysed, the computation takes into account a company's funding currency. For instance, if a company is holding debt for which half is denominated in EUR and the other half in USD, the average of USLIBOR and EURIBOR is used as a benchmark base rate. Risk-adjusted spreads are, for their part, constructed by combining the following items:

- Credit risk spreads: These spreads are established based on the average spreads observed between corporate bonds and government bonds. Spreads are averaged for each credit rating (e.g. AA or BBB), where lower-rated bonds have a higher spread. The terms of these benchmark spreads should ideally match the weighted-average life of each debt transaction in question. Because this paper looks by necessity at corporate-level annual average interest rates, a term of benchmark spreads of one year is applied as a proxy and, in a few cases, is extended to five years where one year rates are not available.
- Government guarantee: These spreads correspond to the additional spreads that would have otherwise been charged absent government guarantees (explicit or implicit). Accredited credit-rating agencies usually base their standalone credit ratings for firms on corporate performance alone, following which they adjust the ratings to account for additional external factors, including expected government support in case of financial distress. Considering such information, these additional spreads represent the increase in interest rates that would occur absent such government support.

Figure A C.1. Estimation of below market borrowings through the construction of a hypothetical market rate of interest



Source: (OECD, 2021[15]).

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Notes

[1] Market share is defined here as the percentage of a company's total sales (domestically and abroad) to the estimated total sales of the global steel sector – for each year in the study period.

[2] Due to numerous missing values in the estimates of corporate tax concessions in the MAGIC database, including this variable would significantly reduce the sample size. For this reason, the analysis in this section focuses on the combined grant- and BMB-to-asset ratio, which allows for a broad coverage of the known subsidy landscape - capturing the two main instruments - while preserving a larger and more consistent sample across consecutive years.

[3] Such a disaggregated study may become useful once more observations (i.e. more firms and more years) are added to the OECD MAGIC database. A loss of statistical significance is typical for too small or reduced samples (and does not preclude for a relationship between variables to exist). When looking at coefficients of (not statistically significant) estimates, they are consistent with the findings for the whole panel detailed in this section. For example, negative spillovers on a firm's market share for subsidies provided to its competitors are highlighted. Results can be provided by the author on demand.

OECD Science, Technology and Industry Policy Papers

Subsidies and market share in the global steel industry

No. 191

This paper examines how subsidies affect competitive outcomes in the global steel industry. Using firm-level data from the OECD MAGIC database covering 2006–2022, the analysis assesses the relationship between government support and market-share developments across steel producers worldwide. Econometric results indicate that subsidies increase recipients' global market share at the expense of less-subsidised competitors. Firms receiving larger support through cash grants and below-market borrowings tend to gain market share even when they exhibit weaker productivity, cost efficiency and financial performance: subsidies weaken the normal link between firm performance and market-share gains. The analysis also identifies negative spill-overs on competing firms, implying that support granted to some producers can reduce rivals' market shares and discourage investment by unsubsidised firms. The results are consistent across OECD Members and partner economies, although subsidisation levels are significantly higher in the latter and hence the dampening of market signals is even more pronounced there. Overall, the evidence suggests that subsidies contribute to resource misallocation, persistent steel excess capacity and shifts in global competitive positions. These findings highlight the importance of greater transparency in industrial support and stronger international cooperation to reduce distortions and support a more level playing field in the global steel sector.